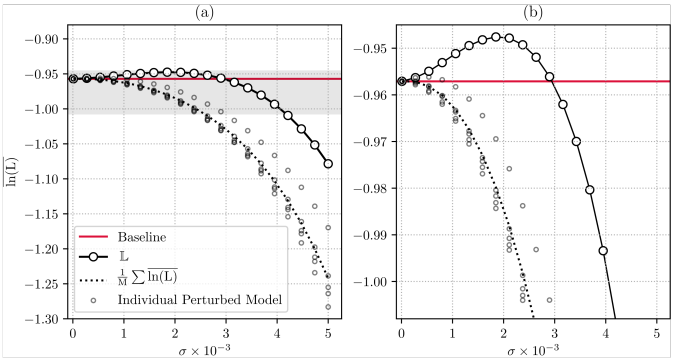


Figure 1: Parameter Ensembling by Perturbation (PEP) on pre-trained InceptionV3 [44]. The rectangle shaded in gray in (a) is shown in greater detail in (b). The average log-likelihood of the ensemble average, $\mathbb{L}(\sigma)$, has a well-defined maximum at $\sigma = 1.85 \times 10^{-3}$. The ensemble also has a noticeable increase in likelihood over the individual ensemble item average log-likelihoods, $\overline{\ln(L)}$ and over their average. In this experiment, an ensemble size of 5 (M = 5) was used for PEP and the experiments were run on 5000 validation images.

We approximate the expectation by a sample average,

$$p(y_i; x_i, \bar{\theta}, \sigma) \approx \frac{1}{m} \sum_j p(y_i; x_i, \theta_j) \qquad \text{where} \qquad \theta_{j=1}^m \underset{\text{IID}}{\leftarrow} N(\bar{\theta}, \sigma^2 I), \tag{5}$$

i.e., the predictions are made by averaging over the predictions of an ensemble. The log-likelihood of the ensemble prediction as a function of $\sigma$ is then

$$\mathbb{L}(\sigma) \doteq \sum_i \ln \frac{1}{m} \sum_j L_i(\theta_j) \qquad \text{where} \qquad \theta_{j=1}^m \underset{\text{IID}}{\leftarrow} N(\bar{\theta}, \sigma^2 I) \tag{6}$$

(dependence on $\bar{\theta}$ is suppressed for clarity). Throughout most of paper we will use $i$ to index data items, $j$ to index ensemble of parameters, and $m$ to indicate the size of the ensemble. We estimate the model parameters as follows. First we optimize $\theta$ with $\sigma$ fixed at zero using a training data set (when $\sigma \to 0$ the $\theta_j \to \bar{\theta}$), then

$$\theta^* = \underset{\bar{\theta}}{\operatorname{argmax}} \sum_i \ln p(y_i; x_i, \bar{\theta}) \ , \tag{7}$$

which is equivalent to maximum likelihood parameter estimation of the base model. Next we optimize over $\sigma$, (using a validation data set), with $\theta$ fixed at the previous estimate, $\theta^*$,

$$\sigma^* = \underset{\sigma}{\operatorname{argmax}} \sum_i \ln \frac{1}{m} \sum_{\theta_j} p(y_i; x_i, \theta_j) \ \text{ where } \ \theta_{j=1}^m \underset{\text{IID}}{\leftarrow} N(\theta^*, \sigma^2 I) \ . \tag{8}$$

Then at test time the ensemble prediction is

$$p(y_i; x_i, \theta^*, \sigma^*) \approx \frac{1}{m} \sum_{\theta_j} p(y_i; x_i, \theta_j) \ \text{ where } \ \theta_{j=1}^m \underset{\text{IID}}{\leftarrow} N(\theta^*, {\sigma^*}^2 I) \ . \tag{9}$$

In our experiments, perhaps somewhat surprisingly, $\mathbb{L}(\sigma)$ has a well-defined maximum away from $\sigma = 0$ (which corresponds to the baseline model). As $\sigma$ grows from 0, $\mathbb{L}(\sigma)$ rises to a well-defined peak value, then falls dramatically (Figure 1). Conveniently, the calibration quality tends to grow favorably until the $\mathbb{L}(\sigma)$ peak is reached. It may be that $\mathbb{L}(\sigma)$ initially grows because the classifiers corresponding to the ensemble parameters remain accurate, and the ensemble performs better as the classifiers become more independent [6]. Figure 1 shows $\mathbb{L}(\sigma)$ for experiments with InceptionV3 [44], along with the average log-likelihoods ($\overline{\ln(L)}$) of the individual ensemble members. Note that in the figures, in the current machine learning style, we have used averaged log-likelihoods, while in

this section we use the estimation literature convention that log-likelihoods are summed rather than averaged. We can see that for several members, $\overline{\ln(L)}$ grows somewhat initially, this indicates that the optimal parameter from training is not optimal for the validation data. Interestingly, the ensemble has a more robust increase, which persists over scale substantially longer than for the individual networks. We have observed this $\mathbb{L}(\sigma)$ "increase to peak" phenomenon in many experiments with a wide variety of networks.

### 2.3 Local Analysis

In this section, we analyze the nature of the PEP effect in the neighborhood of $\theta^*$. Returning to the log-likelihood of a PEP ensemble (Eq. 6), and "undoing" the approximation by sample average,

$$\mathbb{L}(\sigma) \approx \sum_i \ln \mathbb{E}_{\theta \sim N(\theta^*, \sigma^2 I)} \left[ L_i(\theta) \right] \ . \tag{10}$$

Next, we develop a local approximation to the expected value of the log-likelihood. The following formula is derived in the Appendix (Eq 5) using a second-order Taylor expansion about the mean.

For $x \sim N(\mu, \Sigma)$

$$\mathbb{E}_x \left[ f(x) \right] \approx f(\mu) + \frac{1}{2} T_R(Hf(\mu)\Sigma) \ , \tag{11}$$

where $Hf(x)$ is the Hessian of $f(x)$ and $T_R$ is the trace. In the special case that $\Sigma = \sigma^2 I$,

$$\mathbb{E}_x \left[ f(x) \right] \approx f(\mu) + \frac{\sigma^2}{2} \triangle f(\mu) \tag{12}$$

where $\triangle$ is the Laplacian, or mean curvature. The appendix shows that the third Taylor term vanishes due to Gaussian properties, so that the approximation residual is $O(\sigma^4 \partial^4 f(\mu))$ where $\partial^4$ is a specific fourth derivative operator.

Applying this to the log-likelihood in Eq. 10 yields

$$\mathbb{L}(\sigma) \approx \sum_i \ln \left[ L_i(\theta^*) + \frac{\sigma^2}{2} \triangle L_i(\theta^*) \right] \approx \sum_i \left[ \ln L_i(\theta^*) + \frac{\sigma^2}{2} \frac{\triangle L_i(\theta^*)}{L_i(\theta^*)} \right] \tag{13}$$

(to first order), or

$$\mathbb{L}(\sigma) \approx \mathcal{L}(\theta^*) + B_\sigma(\theta^*) \ , \tag{14}$$

where $\mathcal{L}(\theta)$ is the log-likelihood of the base model (Eq. 2) and

$$B_\sigma(\theta) \doteq \frac{\sigma^2}{2} \sum_i \frac{\triangle L_i(\theta)}{L_i(\theta)} \tag{15}$$

is the "PEP effect." Note that its value may be dominated by data items that have low likelihood, perhaps because they are difficult cases, or incorrectly labeled. Next we establish a relationship between the PEP effect and the Laplacian of the log-likelihood of the base model. From Appendix (Eq 34) ,

$$\triangle \mathcal{L}(\theta) = \sum_i \left[ \frac{\triangle L_i(\theta)}{L_i(\theta)} - (\nabla \ln L_i(\theta))^2 \right] \tag{16}$$

(here the square in the second term on the right is the dot product of two gradients) Then

$$\triangle \mathcal{L}(\theta) = \frac{2}{\sigma^2} B_\sigma(\theta) - \sum_i (\nabla \ln L_i(\theta))^2 \tag{17}$$

or

$$B_\sigma(\theta) = \frac{\sigma^2}{2} \left[ \triangle \mathcal{L}(\theta) + \sum_i (\nabla \ln L_i(\theta))^2 \right] \ . \tag{18}$$

The empirical Fisher information (FI) is defined in terms of the outer product of gradients as

$$\widetilde{F}(\theta) \doteq \sum_i \nabla \ln L_i(\theta) \nabla \ln L_i(\theta)^T \tag{19}$$

(see [25]) . So, the second term above in Eq. 18 is the trace of the empirical FI. Then finally the PEP effect can be expressed as

$$B_\sigma(\theta) = \frac{\sigma^2}{2}\left[\triangle\mathcal{L}(\theta) + T_R(\widetilde{F}(\theta))\right] . \tag{20}$$

The first term of the PEP effect in Eq. 20, the mean curvature of the log-likelihood, can be positive or negative, (we expect it to be negative near the mode), while the second term, the trace of the empirical Fisher information, is non-negative. As the sum of squared gradients, we may expect the second term to grow as $\theta$ moves away from the mode.

The first term may also be seen as a (negative) trace of an empirical FI. If the sum is converted to an average it approximates an expectation that is equal to the negative of the trace of the Hessian form of the FI, while the second term is the trace of a different empirical FI. Empirical FI are said to be most accurate at the mode of the log-likelihood [25]. So, if $\theta^*$ is close to the log-likelihood mode on the new data, we may expect the terms to cancel. If $\theta^*$ is farther from the log-likelihood mode on the new data, they may no longer cancel.

Next, we discuss two cases, in both we examine the log-likelihood of the validation data, $\mathcal{L}(\theta)$, at $\theta^*$, the result of optimization on the training data. In general, $\theta^*$ will not coincide with the mode of the log-likelihood of the validation data. **Case 1:** $\theta^*$ is 'close' to the mode of the validation data, so we expect the mean curvature to be negative. **Case 2**: $\theta^*$ is 'not close' to the mode of the validation data, so the mean curvature may be positive. We conjecture that case 1 characterizes the likelihood landscape on new data when the baseline model is not overfitted, and that case 2 is characteristic of an overfitted model (where, empirically, we observe positive PEP effect).

As these are local characterizations, they are only valid near $\theta^*$. While the analysis may predict PEP effect for small $\sigma$, as it grows, and the $\theta_j$ move farther from the mode, the log-likelihood will inevitably decrease dramatically (and there will be a peak value between the two regimes).

There has been a lot of work recently concerning the curvature properties of the log-likelihood landscape. Gorbani et al. point out that "Hessian of training loss ... is crucial in determining many behaviors of neural networks"; they provide tools to analyze the Hessian spectrum and point out characteristics associated with networks trained with BN [9]. Sagun et al. [41] show that there is a 'bulk' of zero valued eigenvalues of the Hessian that can be used to analyze overparameterization, and in a related paper discuss implications that "shed light on the geometry of high-dimensional and non-convex spaces in modern applications" [42]. Goodfellow et al. [11] report on experiments that characterize the loss landscape by interpolating among parameter values, either from the initial to final values or between different local minima. Some of these demonstrate convexity of the loss function along the line segment, and they suggest that the optimization problems are less difficult than previously thought. Fort et al. [7] analyze Deep Ensembles from the perspective of the loss landscape, discussing multiple modes and associated connectors among them. While the entire Hessian spectrum is of interest, some insights may be gained from the avenues to characterizing the mean curvature that PEP provides.

## 3 Experiments

This section reports performance of PEP, and compares it to temperature scaling [12], MCD [8], and Deep Ensembles [26], as appropriate. The first set of results are on ImageNet pre-trained networks where the only comparison is with temperature scaling (no training of the baselines was carried out so MCD and Deep Ensembles were not evaluated). Then we report performance on smaller networks, MNIST and CIFAR-10, where we compare to MCD and Deep Ensembles as well. We also show that the PEP effect is strongly related to the degree of overfitting of the baseline networks.

**Evaluation metrics:** Model calibration was evaluated with negative log-likelihood (NLL), Brier score [3] and reliability diagrams [34]. NLL and Brier score are proper scoring rules that are commonly used for measuring the quality of classification uncertainty [36, 26, 8, 12]. Reliability diagrams plot expected accuracy as a function of class probability (confidence), and perfect calibration is achieved when confidence (x-axis) matches expected accuracy (y-axis) exactly [34, 12]. Expected Calibration Error (ECE) is used to summarize the results of the reliability diagram. Details of evaluation metrics are given in the Supplementary Material (Appendix B).